

Build Your AWS Cost Control Centre

BaseStack Academy · AWS Cloud Accelerator

Time

60 min

Objective

Set up real cost controls in your AWS account, explore every pricing model, and use the Pricing Calculator to build a cost estimate for a Nigerian startup scenario.

Prerequisite

AWS Free Tier account

 **Portfolio Goal** — Screenshot your completed AWS Budgets alert + Pricing Calculator estimate and post to LinkedIn. Tag it **#AWSCostControl** **#BaseStackAcademy**

Lab Steps

Step 1 Activate Your Cost Safety Net — AWS Budgets

The instructor said it, and it is worth repeating: the very first thing you do on a new AWS account is set a \$0 budget alert. Do it now.

- 1a. In the AWS Console search bar, type Budgets and click AWS Budgets.
- 1b. Click Create budget → Use a template → Zero spend budget.
- 1c. Enter your email address for notifications. Click Create budget.
- 1d. In your tracker, document: Budget name · Alert threshold · Email set.



If AWS sends you a cost alert later in this course, it means you left a resource running. Stop the resource immediately, then check the billing console.

Step 2 Explore the Free Tier Dashboard

Navigate to your AWS Free Tier usage dashboard and answer the following questions in your tracker.

5. In the search bar type Free Tier, then open the AWS Free Tier page.
6. Locate EC2, S3, and Lambda in the service list.

Service	Free Tier Type	Monthly Limit
EC2 (t2.micro / t3.micro)	12-month free	750 hours / month

S3	12-month free	5 GB storage · 20K GET · 2K PUT
Lambda	Always free	1M requests / month

7. Add a Free Tier Type column to your services tracker and fill in the three rows above.

Step 3 Compare All Four Pricing Models in the EC2 Console

You will look at real pricing data — without launching any instance.

8. Search for EC2 → On the left sidebar click Instances → Launch Instances (stop at the first screen).
9. Notice the instance type dropdown. Select t3.micro. This is the On-Demand baseline.
10. Now go to: aws.amazon.com/ec2/pricing/on-demand — find the t3.micro price for us-east-1.
11. Go to: aws.amazon.com/ec2/pricing/reserved-instances/pricing — find the 1-year No Upfront price for t3.micro.
12. Go to: aws.amazon.com/ec2/spot/instance-advisor — search t3.micro in us-east-1. Record the Spot discount %.

Document your findings in this table:

Pricing Model	Price / hr (t3.micro)	Commitment	Best For
On-Demand	\$_____	None	Unpredictable load
Reserved (1-yr, No Upfront)	\$_____	1 year	Steady workloads
Spot (t3.micro, us-east-1)	~\$_____ (varies)	None	Fault-tolerant batch
Free Tier (t2.micro)	\$0.00 (up to 750 hrs)	12 months	Learning & labs

Step 4 Build a Startup Cost Estimate — AWS Pricing Calculator

You are the cloud architect for a Lagos fintech startup. Build their 3-month AWS cost estimate.

13. Go to calculator.aws and click Create estimate.
14. Add EC2: 1 × t3.micro, On-Demand, Linux, us-east-1, 730 hrs/month.

15. Add S3: Standard storage, 10 GB/month, 50,000 GET requests, 5,000 PUT requests.
16. Add RDS: db.t3.micro, MySQL, Single-AZ, 20 GB General Purpose SSD.
17. Review the monthly total. Click Save and share → Copy public link.
18. Now change EC2 to Reserved Instance (1-year, No Upfront). What is the new monthly total?

Configuration	Monthly Estimate
EC2 (On-Demand) + S3 + RDS	\$ _____
EC2 (Reserved 1-yr) + S3 + RDS	\$ _____
Savings with Reserved (3-year projection)	\$ _____

Step 5 Tour AWS Cost Explorer

Cost Explorer is where you review real spending after it happens. Activate it so it is ready when your account has usage data.

19. In the search bar type Cost Explorer and open it.
20. If prompted, click Enable Cost Explorer (free to activate, free to use).
21. Explore the Date Range and Group By filters. On a new account the graph will be empty — that is fine.
22. In your tracker, note: what two Group By options would most help a startup track costs?

Screenshot Checklist

AWS Budgets dashboard showing your \$0 Zero Spend Budget	☐
Pricing Calculator estimate showing On-Demand vs Reserved monthly totals	☐
Your completed pricing model comparison table (Step 3)	☐
Cost Explorer activated (even if graph is empty)	☐

Portfolio Post — Copy & Customise

'Just set up my AWS cost controls: \$0 budget alert  Free Tier tracker  Pricing Calculator estimate  On-Demand vs Reserved comparison . The difference in a 3-year Reserved vs On-Demand bill is wild. Here's my estimate: [paste your calculator link]. #AWSCloudArchitect #CloudEconomics #BaseStackAcademy'



Bonus Challenge — Push Your Skills Further

Scenario: Paystack Black Friday Surge

Paystack expects 10× normal traffic on Black Friday (1 day only). Normal load = 2 × t3.medium EC2 instances.

- Task 1 — In the Pricing Calculator, model: 2 × t3.medium (Reserved, 1-yr) for baseline + 18 × t3.medium (On-Demand) for the 24-hour Black Friday surge.
- Task 2 — Now replace the 18 On-Demand surge instances with Spot Instances. Estimate the cost difference.
- Task 3 — Write one paragraph explaining which model you would recommend for the surge, and why Spot may not be appropriate for Paystack's payment processing core.